

EUNCHEOL IM

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PERSONAL INFORMATION

- Date of Birth : Oct 12th, 1995
- Place of Birth : Jeju island, Republic of Korea
- Nationality : Republic of Korea

EDUCATION

Korea University Ph.D. Student in School of Electrical Engineering Advisor : Myo-Tag Lim GPA: 4.2 / 4.5	March 2023 - Present Seoul, Republic of Korea
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Current Position : Student Researcher
Affiliation : Center for Humanoid Research, Korea Institute of Science and Technology (KIST)
PI : Dr.Yisoo Lee

Jeju National University M.S. in Mechanical Engineering Thesis: "Unmanned Vehicle Mobility Platform of Underwater Pole-Climbing robot" Advisor : Jin Hwan Ko GPA: 4.3 / 4.5	March 2020 - Feb 2022 Jeju, Republic of Korea
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Jeju National University B.S. in Mechanical Engineering Advisor : Jin Hwan Ko GPA: 4.35 / 4.5 Summa Cum Laude	March 2014 - Feb 2020 Jeju, Republic of Korea
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SKILLS AND INTERESTS

Interests	Robotics, Optimal Control, Loco-manipulation, Manipulation, Bi-manipulation
Skills	Auto CAD, SolidWorks, Inventor Python, C/C++, CUDA qpOASES MuJoCo, Issac-sim, Issac-gym, Isaac-Lab

AWARDS AND HONORS

Academic Excellence Scholarship, Jeju National University, Korea	Mar. 2018 - Aug. 2019
Jeju Talent Development Scholar, Jeju Lifelong Education and Scholarship Promotion Institute, Korea	May. 2020
Future Agricultural Talent Scholar, NongHyup Foundation, Korea	Jun. 2021 - Dec. 2021
Excellence in Technology Award, Korea Institute of Science and Technology (KIST) AI-Robot Research Center, Korea	Dec. 2024

RESEARCH EXPERIENCE

Research Student - Underwater Robot Demonstration Center, Korea Institute Of Ocean Science and Technology, Korea	Feb. 2020 - Feb. 2020
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- Experiment to find out the tendency of flow velocity and flow velocity distribution of circulation

Research Intern - Underwater Robot Demonstration Center, Korea Institute Of Ocean Science and Technology, Korea Aug. 2020 - Aug. 2020

- Indoor Positioning System (Vision Tracking) Calibration, Error Cause Identification, and Analysis

Research Intern - Smart Mobility Systems Research Center, Smart Mobility R&D Division, Korea Institute of Robotics and Technology Convergence (KIRO), Korea Jan. 2021 - Feb. 2021

- Design and Test of Mobile platform support structure to prevent trajectory departure

Research Intern - Smart Mobility Systems Research Center, Smart Mobility R&D Division, Korea Institute of Robotics and Technology Convergence (KIRO), Korea Jul. 2021 - Aug. 2021

- Assemble and Test Mobile Platform and attend to robot control experiment process

Research Intern - Center for Intelligent and Interactive Robotics, Korea Institute of Science and Technology (KIST), Korea Mar. 2022 - Feb. 2023

- Design of Dual-arm robot controller and contact modelling

PROJECTS

Turtle-Mimetic Tidal Generator

National Research Foundation of Korea (NRF), Korea

Sep. 2020 - Sep. 2023

- Design of Buoyancy Control Device and generator frame

Core Technologies for Metabot Platform

Institutional Proprietary Research Project, Korea

Jan. 2023 - Dec. 2024

- Development of Dual-arm controller

Real-Time Whole-Body Model Predictive Control for Multi-Joint Mobile Systems in Dynamic Environments

National Research Foundation of Korea (NRF), Korea

Apr. 2024 - Present

- Development of Model Predictive Control Framework

Development of Full-body Control-Based Dual-Arm Mobility and Manipulation Technology

Hyundai Motor Company, Korea

Oct. 2024 - Present

- Development of Dual-arm controller using Model Predictive Control

Development of Robust Autonomous Mobility and Task Execution Technology for Humanoids

Institutional Proprietary Research Project, Korea

Jan. 2025 - Present

RESEARCH PUBLICATION(SCIE)

Jinho Kim, **Euncheol Im**, Yisoo Lee[†], Youngsu Cha[†]” *Quadrupedal robot with tendon-driven origami legs*”, Sensors and Actuators A: Physical (2024)

Dongwhan Kim, **Euncheol Im**(co-first author), Yujin Kim, Myo-tag Lim, Yisoo Lee[†]” *Single-Instance Sampling for Computationally Efficient and Accurate Real-Time Task Space MPPI Control*”, Transactions on Robotics (2025)

Yunseoung Na, **Euncheol Im**, Sujin Lee, Yisoo Lee, Sungwook Yang, Yonghwan Oh, Yong Seok Ihn[†]” *REIST: A Novel Robotic Wrist with Reversed-Envelope Characteristics for Manipulation in Confined Spaces*”, IEEE/ASME Transactions on Mechatronics (2026)

CONFERENCE

Euncheol Im, JinHwan Ko[†]” *Experimental Study on a Mobility Platform of Underwater Pole-Climbing Robot for Offshore Wind Turbine Substructure*”, International Conference on Control, Automation and Systems (ICCAS 2021), Jeju, Korea (Oct. 2021) (Poster presentation)

Euncheol Im, HyeonMin Sim, JinHwan Ko[†]” *A wheeled underwater pole climbing robot of dual configuration for maintenance of offshore wind turbine substructures*”, International Conference on Ubiquitous Robots (UR 2022), Jeju (Jun. 2022), Korea (Poster presentation)

Jaesung Ahn, **Euncheol Im**, Yisoo Lee[†]” *Learning Impulse-Reduced Gait for Quadruped Robot using CMA-ES*”, International Conference on Ubiquitous Robots (UR 2023), Hawaii (Jun. 2023), United State (Oral presentation)

Gunoo Park, Jaewan Bak, Yunsoo Seo, **Euncheol Im**, Hoseok Lee, Jongbok Lee, Jongwon Lee, Yisoo Lee[†]” *Real-Time Model Predictive Control of Nonlinear Coupled Joints Using MPPI: Application to Humanoid Ankle Joints*”, International Conference on Robotics & Automation (ICRA 2026), Vienna (Jun. 2026), Austria (Poster presentation)

Euncheol Im, Taehyun Kim, Yonghwan Oh, Myotaeg Lim, Yisoo Lee[†]” *Analytic Torque-Sampling MPPI: A GPU-Accelerated Framework for Real-Time Manipulation*”, International Conference on Intelligent Robots and Systems (IROS 2026), Pittsburgh (Sep. 2026), USA (Oral-underreview)

Yunsoo Seo, Sol Choi, **Euncheol Im** Myotaeg Lim, Yisoo Lee[†]” *RGB: RL Guided Whole-Body MPPI for Humanoid Control* ”, International Conference on Intelligent Robots and Systems (IROS 2026), Pittsburgh (Sep. 2026), USA (Oral-underreview)

PUBLICATION (DOMESTIC)

Euncheol Im, JinHwan Ko[†]” *An Experimental Study on an Underwater Column-Climbing Robot for Maintenance of Offshore Wind Turbine Support Structures*”, Korea Robotics Society (KROS 2022)